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Research Paper

The Effect of Boundary-Spanning Search on Breakthrough Innovations of New Technology Ventures

CHEN-JU LIN* & CI-RONG LI**

**Department of Accounting Information, Tzu Chi College of Technology, 880, Sec.2, Chien-Kuo Rd. Hualien City, 970, Taiwan, ROC, **Faculty of Management, Fuqing Branch of Fujian Normal University, No.1, Xiaoyuan Quarter, Longjiang, Fuqing City, 350300, Fujian*

ABSTRACT This study expands the scope of research on knowledge search and innovation search by investigating the effect of boundary-spanning search behavior on breakthrough innovations of new ventures. Based on an environmental coalignment perspective, we argue that boundary-spanning search is a key activity for new ventures so that they can deal with different market forces to enhance their breakthrough innovations. Most of the hypotheses proposed are supported by the results of a survey of 227 ventures in the high-technology zone of Shenzhen. We found that boundary-spanning search is not only associated positively with breakthrough innovations of new ventures but also plays a significant role in influencing the effect of market forces on both types of innovations. This paper offers insight into the way in which new ventures can achieve both market innovations and technological innovations through different types of boundary-spanning search. Furthermore, this study argues that boundary-spanning search is an important tool that enables new ventures to create more innovative performance under competitive environments.

KEY WORDS: boundary-spanning search, environment-performance paradigm, breakthrough innovation, transitional economies, new ventures, exploration and exploitation

1. Introduction

Responding to rapid technological changes and shortened product cycles, firms need to *search* for and access knowledge, assimilate it, and apply it to commercial ends. Studies on knowledge search or innovation search recognize two types of innovation search: (i) local search or exploitation, which searches for solutions in the neighborhood of its current expertise and (ii) distant search or exploration, which searches for solutions besides what the firm already knows. Recent work indicates that, facing time pressure, changing circumstances, and resource scarcity, firms need to exercise successful innovation

Correspondence Address: Chen-Ju Lin, Assistant Professor, Department of Accounting Information, Tzu Chi College of Technology, 880, Sec.2, Chien-kuo Rd. Hualien City, 970, Taiwan. Tel.: 886-3-8572158 ext. 632. Email: jjlin0208@gmail.com

searches to facilitate the expansion of their organizational boundaries and go beyond their existing knowledge space (Rosenkopf & Nerkar, 2001; Greve, 2003; Laursen, 2012).

A growing literature is beginning to expand our understanding of the different ways in which firms can engage in searching knowledge across their boundaries (Rosenkopf & Nerkar, 2001; Katila, 2002; Sidhu *et al.*, 2007; Katila & Chen, 2008). Different types of distant searches may promote innovativeness and firm performance in different ways, even though previous research on the implications of market forces or industrial structures for innovation search is scant. Katila and Chen (2008) also note the need for future research to consider the influences of competitive environments in which firms encounter different constraints on and opportunities for innovation and distant search. Hence, a question of both theoretical and practical interests is whether firms facing competitive environments can develop innovation through engaging in boundary-spanning search.

In line with literature on market forces (Li & Calantone, 1998; Voss & Voss, 2000), this paper examines the three most fundamental characteristics of market forces in a competitive environment: demand uncertainty, technology uncertainty and competitive intensity. These three characteristics represent the influences of customers, technology and competitors in industrial structures. In general, market forces have more influence on new ventures using innovative strategies for ensuring success (Li & Atuahene-Gima, 2001) than they do on long-standing, large-scale and established firms. In contrast to established firms, new ventures have not yet inherited an abundance of technological and marketing competencies that came from traditional and fundamental experience gained through successive innovations. Limited resources force new ventures to be more vulnerable than established firms. Due to an insufficient knowledge base, they may be exposed to severe defeats when they engage in local search or self-exploitation for incremental innovations. Hence, another question is whether new ventures should engage in distant searches or boundary-spanning searches to leverage their innovative capabilities, in order to capture innovation performance in a competitive environment.

This paper adopts the environmental strategy coalignment perspective as one way of clarifying the relationship among external market forces, different types of distant search and breakthrough innovations for new ventures. The core premise of the environmental strategy coalignment perspective is that the *fit* between strategy and external environment has significant positive implications for market performance (Van de Ven & Drazin, 1985; Venkatraman & Prescott, 1990; Volberda *et al.*, 2012). Accordingly, to respond to the different forces of the external environment, ventures may need to adopt *appropriate* types of boundary-spanning search to reconcile the gloomy implication of uncertainty and competition for innovativeness. With this perspective, we argue that new ventures should capture the opportunities from external market forces to develop innovations when boundary-spanning searches deal with the specific requirements of market forces.

2. Theoretical Framework and Hypotheses

One aspect of the theoretical logic of the environmental strategy coalignment perspective is that as a firm aligns its resource deployment with the requirements of external environment, its performance becomes significantly better than others that did not achieve coalignment (Venkatraman & Prescott, 1990; Zajac *et al.*, 2000; Lukas *et al.*, 2001). When different market forces cause different constraints and opportunities (Hrebiniak & Joyce, 1985;

Sidhu *et al.*, 2007; Yang & Li, 2011) ventures should adopt distinct types of boundary-spanning search that provide different knowledge and information to respond to various sources of market forces. Another aspect of the theoretical logic points out that a strategy corresponding with the external environment can contribute to better performance (Hambrick, 1983; Hambrick & Lei, 1985). Different types of boundary-spanning search can seize the opportunities and mitigate the uncertainties from different sources of market force. Thus, this study proposes that if new ventures engage in appropriate boundary-spanning search strategies that can deal with the *specific* requirements of the market forces, they can capture innovation performance. This implies that different types of boundary-spanning search may influence the effect of market force on new venture innovation performance.

In this study, we have broken boundary-spanning search into three categories: customer-driven (demand-side) search, technology-driven (supply-side) search and competitor-driven (spatial-side) search. By definition, supply search focuses on seeking new knowledge about technology as well as the organizational aspects that bear on the input–output conversion, whereas demand search emphasizes devoting efforts to the discovery of new insights in its own and allied markets (Sidhu *et al.*, 2007). Spatial search focuses on finding opportunities, localized know-how and operational experience in different regimes including international regions and extra-industrial regions (Katila, 2002; Phene *et al.*, 2006).

Ahuja and Lampert (2001) contends that when facing dynamic and competitive environments, firms are more likely to experience the effect of restrictions in creating continuous or incremental innovations that are limited in being able to guarantee firm success or survival. Because new ventures lack mature organizational processes and ready technologies, reliance on incremental innovations is at a competitive disadvantage against large and established firms (Lee *et al.*, 2001; Atuahene-Gima *et al.*, 2006). The survival of new ventures facing turbulent environments depends largely on breakthrough innovations that serve to introduce novel and differentiated products, processes and marketing innovativeness.

Breakthrough innovation, in line with a prior work (Wind & Mahajan, 1997), is defined as novel, unique or state-of-the-art technological advances as well as the capability to alter the venture's competitive pattern in the market. More recently, research for breakthrough innovations is divided into two dimensions: tech-based and market-based innovations (Christensen & Bower, 1996; Benner & Tushman, 2003). *Tech-based innovation* refers to the adaptation of new and advanced technologies and amelioration of customer benefits, both of which are relative to existing products for customers in existing markets. *Market-based innovation* is a departure from serving the existing mainstream markets and involves new and different technologies, creates a set of fundamental knowledge and generates new customer values for emerging markets. New ventures tend to pursue both tech-based and market-based innovations under the conditions of demand uncertainty and technological turbulence, whereas new ventures facing an intensive competitive market tend to undertake market-based innovations rather than tech-based innovations (Zhou *et al.*, 2005).

To answer the two questions above, this study builds a research framework (see Figure 1) and provides three explanations: (1) the link between different forces of environmental dynamics (technology turbulence, market turbulence and competitive intensity) and different types of boundary-spanning search (supply-side search, demand-side search and spatial-side search); (2) the effect of boundary-spanning search behavior on breakthrough innovations (tech-based innovations and market-based innovations) and (3) how boundary-spanning search behavior influences the effect of market forces on breakthrough innovations.

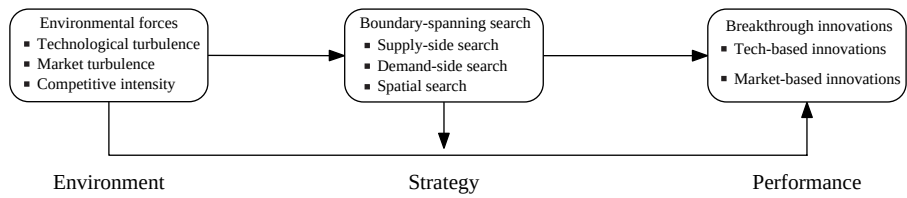


Figure 1. Theoretical model.

2.1 The Effects of External Environmental Forces on Boundary-Spanning Search

When a firm encounters a technologically dynamic environment, it may face rapid technological advances, while the value and impact of existing technology deteriorates very quickly (Sirinivasan *et al.*, 2002), especially in China (Gao *et al.*, 2007). Technology-oriented external knowledge is vital to firms to overcome dynamic and complex technological market forces (Powell *et al.*, 1996; Shan & Song, 1997) because most of the existing technology is limited to guarantee any competitive advantage. Boundary-spanning supply search that can provide new knowledge through searching in external technological and other supply-side domains adds variation in the technological evolution repertoire to increase the chances of finding new combinations (Katila & Ahuja, 2002; Phene *et al.*, 2006). Thus, ventures encountering technological turbulence may engage in supply-side search to experiment with new technology and reduce uncertainty through technological newness. Therefore, we propose:

H1: When experiencing a higher level of technological turbulence, new ventures are more likely to pursue supply-side distant search.

As the market demand becomes more dynamic, unstable new ventures may experience munificence in demand (Atuahene-Gima *et al.*, 2006) and face challenges to grasp customers' *real* needs. Knowledge regarding changes in customers' preferences and expectations is vital for the firm to sustain its competitive advantage. Boundary-spanning demand search, similar to proactive market orientation or exploration (Atuahene-Gima *et al.*, 2005; Siduh *et al.*, 2007), can provide external knowledge through scanning changes in customer preferences and new sets of potential customers, identifying other ways to satisfy customer needs and searching market niches unoccupied by rivals. Established firms typically tend to maintain links with existing customers and suppliers, which prevent them from engaging in demand-side distant search. In contrast, new ventures have greater flexibility to track changes in customer preferences and respond in a timely manner, which are not limited by established links with customers and suppliers. Hence, we propose:

H2: When experiencing higher market turbulence, new ventures are more likely to pursue demand-side distant search.

Competitive intensity is characterized by violent price competition, product and service diversity, and heavy inputs, such as advertising (Porter, 1980). When competitive conditions become more demanding, strongly focusing on risky and proactive innovativeness can be more hazardous for new ventures than for the established firms. Also, it can be considerably harmful to the ventures' viability (Miller & Friesen, 1983) and to the firms' confrontations with

intensive pressures for higher efficiency and lower prices (Matusik & Hill, 1998). However, new ventures can fix their eyes on the existing market while established firms may ultimately find it difficult to compete due to the lack of slack resources for exploration and existing competences for exploitation. Therefore, all ventures need to focus on a more varied set of knowledge for the efficient recombination regarding the production, quality improvement and distribution of products to capture competitive advantages. Boundary-spanning spatial search can contribute valuable knowledge by gaining penetration into the new-to-the-industry regions in foreign markets or extra-industries. Such knowledge may be old for some regions, but can provide low-risk and low-cost new insights which contribute novelty with cost-effectiveness and efficiency for the ventures (Katila, 2002; Phene *et al.*, 2006); thereby, such knowledge may provide a competitive advantage to competitors whose focus on search efforts is more localized. Hence, we propose:

- H3:* When experiencing higher competitive intensity, new ventures are more likely to pursue spatial-side distant search.

2.2 The Effects of Boundary-Spanning Search on Breakthrough Innovations

The advantages of the supply-side search in collecting new insights about technological aspects and the operational processes (Sidhu *et al.*, 2007) can add a new technological variant into the problem-solving repertoire (Katila & Ahuja, 2002) that prompts ventures into finding valuable new innovativeness for technological advances. The acquired knowledge also adds to the odds of promising tech-based innovations through increasing the venture's ability to sense technological novelty that is conducive to grasp a sudden technology evolution (Rosenkopf & Nerkar, 2001). New ventures engaging in supply-side search can obtain advancements regarding organizational operation and acquire varied sets of new technological knowledge that provide valuable resources for changing the technological trajectory. However, the external technological knowledge may not value market-based innovations, because such knowledge may be too technologically straightforward and are useless in finding new or emerging markets and offering new solutions to unexpressed needs. In addition, new ventures aggressively engage in supply-side search that is unlikely to invest substantially in seeking and using market knowledge to create value in new segments (Christensen, 1997). Thus,

- H4:* The greater degree new ventures pursue supply-side search, (a) the more likely new ventures are to develop tech-based innovations and (b) the less likely they are to develop market-based innovations.

Engaging in demand-side search sharpens a new venture's awareness and knowledge about the needs of current and potential customers, offers insights about customer product use and substitutive patterns and gains new-to-firm, customer-driven ways to the product offering (Day, 1994). To grasp niches in the oncoming market, ventures engaging in demand-side search can grasp the value of the new segment and potential customer preferences, both of which prompt them to find novel and useful ways to satisfy emergent needs. In addition, demand-side search can facilitate new ventures to quickly recognize the existing competitors' superiority or inferiority that relate to technologies and marketing activities. Thus, new ventures can grasp information about customer reception for existing

products, their pattern of product use and competitive prices of customer products (Sidhu *et al.*, 2007). New ventures can use the knowledge collected from competitors and allied markets to address the needs of existing markets and provide greater customer benefits than do the existing product (Chandy & Tellis, 1998). The external market knowledge also provides valuable information about the direction of potential technological trajectory that prompts ventures to capture state-of-the-art technological advances (Benner & Tushman, 2003). Thus,

- H5: The greater degree new ventures pursue demand-side search, (a) the more likely new ventures are to develop tech-based innovations and (b) the more likely they are to develop market-based innovations.

To develop market-based innovation, firms need to focus on the new market segment and new markets (Benner & Tushman, 2003). Spatial distant search providing multi-regional and extra-industry knowledge is beneficial for ventures to efficiently find new insights for recombination that is cost-effective and enhances rapid success in new markets. Such external knowledge may be old for other regions but may contribute to the new segment beneficially for where the industry ventures are located. For example, Toyota successfully entered the car market of the USA by introducing small cars, which used its existing experience of the Japanese market to provide new segment benefits to U.S. customers. Ventures also can rapidly renew their products and technology at lower costs through external searches in different geographical locations (Nelson & Winter, 1982), and in different other industries (Katila, 2002). By using spatial-side search, new ventures grasp the best practice paradigm relating to the organization of production and the improvement of quality outside the industry, then interconnect pieces of knowledge available within the organization to that from the outside, and then they process the recombination of a variety of different *localized* and non-localized knowledge that facilitate ventures to find the way for changing the technological trajectory inside the industry in which they are located (Kogut & Zander, 1992; Henderson & Cockburn, 1994). Thus,

- H6: The greater degree new ventures pursue spatial-side search, (a) the more likely new ventures are to develop tech-based innovations and (b) the more likely they are to develop market-based innovations.

2.3 The Moderating Effects of the Boundary-Spanning Search

Applying supply-side search provides new ventures with new technological knowledge elements that increase their problem-solving repertoire by which new combinations of knowledge or innovation create advance, next-generation products (Nelson & Winter, 1982). When technology is undergoing rapid change in an industry, existing technologies become quickly obsolete and new advanced technologies are in demand. To diminish obsolescence and constraints, new ventures need to accumulate new knowledge for the potential in uncovering new technological areas. The capacity to search more broadly in the technological aspect and other supply-side domains facilitates firms to generate more valuable breakthrough innovations (Fleming & Sorenson, 2004). In contrast, as the technological process of an industry becomes less dynamic, an emphasis on supply-side

search may be useless to develop breakthrough innovations due to the unnecessary costs for novel knowledge acquisition and integration. Thus, we suggest that

- H7:* New ventures encountering a high-technological turbulence have a greater extent of 9 (a) tech-based innovations and (b) market-based innovations when engaging in a higher degree of supply-side search.

When the demand of the market is turbulent, customers' needs change very quickly and firms find it very difficult to satisfy customers by offering favorite products and services with their existing resources and capabilities. Zhou *et al.*, (2005) proposed that unpredictable circumstances may force new ventures to effectively respond to customers by reshaping the customers' behavior and by proactively creating demand to fit the firms' purpose on demand control. Under these conditions, new ventures engaging in boundary-spanning demand search can increase the ability of developing market-based innovations through grasping new insights related to new segment benefits, new customer value, and potential market niches. In addition, when new ventures face demand dynamism, demand-side search can grasp customers' latent and current needs so as to provide valuable knowledge for the future direction of new advanced technologies, which facilitate ventures to invest resources to develop tech-based innovations. Consequently, we suggest:

- H8:* New ventures encountering a high turbulence have a greater extent of (a) tech-based innovations and (b) market-based innovations when engaging in a higher degree of demand-side search.

An intensively competitive market may bring severe price wars and diverse alternatives from newly introduced products. To purposely defeat its competitors, firms usually spend much advertising expense to promote new products and to launch new services (Porter, 1980). To accelerate the speed of introducing new products into the market, firms typically tend to improve and imitate competitors' products and capabilities because the activities of product imitation can cost less than the high cost of product innovation (Day & Wensley, 1988). The tension of developing tech-based innovations may be diminished due to the imitation of competitors' behavior. The spatial-side search facilitates new ventures to acquire useful localized know-how and organizational experience from various regions that are beneficial to the discovery of new insights and a niche in market for themselves. Through searching reliable external knowledge in different regions for recombination, new ventures can rapidly advance with low risk of the existing technologies and discover ways to change the technological trajectory in a cost-effective manner. This information means that by engaging in boundary-spanning spatial search, firms can be in a favorable position to create tech-based innovations for new ventures within a highly competitive market. In addition, as some firms encounter an intensely competitive market, they attempt to seek new opportunities in new markets to compete against existing rivals (Porter, 1980). They take action to identify different value systems in different geographies through the spatial-side search. Through searching different organizational experiences, ventures are prompted by localized know-how to recheck their opportunities and markets which may be undeveloped or unserved. More specifically, focusing on boundary-spanning spatial search can enhance new ventures' ability to discover latent customer needs and identify emergent market niches. Consequently, we propose:

- H9: New ventures encountering a high competitive intensity have greater (a) tech-based innovations and (b) market-based innovations when engaging in a higher degree of spatial-side search.

3 Research Methods

3.1 Sample and Data Collection

The sampling frame of this study consisted of new technology ventures located in Shenzhen, the most developed high-technology industrial zone in China. We drew on a random sample of 750 new ventures (i.e., 8 years or younger) from a mailing list of electronics firms in this high-technology zone. We contacted each firm by phone to solicit its participation in this study, and then a total of 378 ventures agreed to participate in the study.

Our study used an on-site interview approach. A trained interviewer scheduled appointments with two informants (general managers, marketing managers or product development managers) in each venture, presented each of them with a questionnaire and finally collected the completed questionnaire. In China, this approach is better than traditional mail surveys due to quality control and reliability of the data (Gao *et al.*, 2007). Our study obtained data from 227 usable questionnaires for an effective participation rate of 60.0%. The average age of ventures was 4.41 years ($SD = 1.5$ years). Of the responding ventures, 22.4% were in electronics and information technology, 21.5% in computer software and hardware industries, 18.3% in new energy and new materials, 12.6% in integrated optical-mechanical and electrical products, 10.4% in pharmaceutical and biotechnology, 9.6% in automotive components and 5.2% in other areas. Thus, we compared a sample of participating and non-participating ventures in sales, ventures' age and number of employees to examine the response bias. By analyses of variance, the results found no significant differences in sales, ventures' age and number of employees.

3.2 Measures, Reliability and Validity of Constructs

To examine the consistency between the first and the second informants, we first checked the inter-respondent correlation, and the results showed a significantly positive correlation above 0.65. Then, we computed the inter-rater agreement index ($r_{wg(j)}$) for each item (LeBreton *et al.*, 2005). These values ranged from 0.80 to 0.86. The lowest $r_{wg(j)}$ was 0.80 (above the 0.7 benchmark) and indicated a great consistency between the two informants (George, 1990). For these reasons, our study averaged the responses from the two informants to evaluate each measure.

Based on prior studies (Jaworski & Kohli, 1993), we measured market turbulence with a four-item scale ($\alpha = 0.81$) that asked informants to evaluate the perceived speed and magnitude of change in customer demand, product preference and emergence of new customer segments in their industry. We measured technological turbulence with a four-item scale ($\alpha = 0.71$) that asked respondents to assess the perceived magnitude of dynamics in technology, including the variety of new product introductions offered by new technology in their industry. Thus, we measured competitive intensity ($\alpha = 0.72$) with four items that reflected the extent to which a firm's external environments are subjectively interpreted as intensely competitive.

Our study adopted the measure of boundary-spanning search from Sidhu and his colleague's (2007) research. A six-item scale ($\alpha = 0.83$) measured demand-side search that addressed the degree to which a venture devoted efforts to the discovery of new insights regarding market segments, product use and substitution patterns and customer preferences and needs in their markets. We captured supply-side search ($\alpha = 0.81$) with a six-item scale by asking the informants to assess the degree to which ventures dedicated resources to search for new knowledge about technological and organizational routine aspects. A four-item scale ($\alpha = 0.77$) measured spatial search that asked respondents to evaluate the extent to which ventures search for opportunities regional know-how about operation and marketing in different geographic locations.

We adapted a subjective measure for breakthrough innovations due to the lack of archival sources of innovation data in the context of most emerging economies (Atuahene-Gima, 2005). On the basis of Zhou and his colleague's (2005) work, we measured tech-based innovation with four items ($\alpha = 0.78$) that tapped the degree of technological progress and benefit improvement in existing products. Thus, our study measured market-based innovations with four items ($\alpha = 0.73$) based on customer's view that addressed the extent to which mainstream customers have difficulty in evaluating and readily accepting new products.

In this study, several variables were controlled for testing the hypotheses. Our study used venture size as measured by the number of employees in order to control for greater complexity and economies of scale that arise in large firms. Venture ownership was controlled since information-processing routines of new ventures may differ from types of ownership (i.e., state-owned, joint share, privately owned and foreign invested firms). We coded ownership type: state-owned as 0 and the joint share/private owned/foreign invested as 1. We also controlled for slack resources and product development alliances since greater resources reflect the increased search for boundary-spanning information and increased development of breakthrough innovations.

Finally, we conducted a statistical check for common method variance (CMV) with the Harman one-factor method (Podsakoff & Organ, 1986). If CMV is a serious research bias, a single factor should account for most of the variance. A principal components factor analysis of all items yielded 11 factors with eigen values greater than 1.0 and total explained variance of 61%. The results indicated that CMV may not be a serious research bias because the variance accounted by the first factor was only 9.43% (Menon *et al.*, 1999). In addition, all items loaded cleanly on separate factors without any cross-loadings above 0.5. In a confirmatory factor analysis, each measure loaded significantly on the expected constructs, which demonstrates convergent validity. Together, the model fit indexes ($\chi^2 = 1048.08$ $df = 944$, $RMR = 0.03$, $CFI = 0.96$, $GFI = 0.85$ and $RMSEA = 0.02$) suggest that the model fit is acceptable. In Table 1, the diagonal elements represented the square roots of the average variance extracted (AVE) for each of the constructs are greater than the off-diagonal elements and satisfy the criterion of discriminant validity (Fornell & Larcker, 1981). Finally, the constructs' (α previously reported and the composite reliabilities (CRs) presented in Table 1 indicate that each exceeded the accepted reliability threshold of 0.60. We averaged the measured score of each construct for further analysis.

Table 1. Correlation matrix and descriptive statistics of measures.

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10	11	12
(1) Tech-based innovation	4.54	0.57	0.68											
(2) Market-based innovation	4.69	0.51	0.12	0.63										
(3) Demand dynamism	4.66	0.60	0.00	0.20	0.72									
(4) Technology turbulence	4.90	0.49	0.08	0.06	0.03	0.62								
(5) Competitor intensity	4.89	0.50	-0.23	0.06	0.08	0.03	0.62							
(6) Demand-side search	4.60	0.53	0.14	0.27	0.16	0.07	-0.09	0.68						
(7) Supply-side search	4.51	0.50	0.29	-0.11	0.15	0.14	-0.05	-0.04	0.65					
(8) Spatial-side search	4.71	0.53	0.18	0.19	-0.03	0.02	0.14	0.00	0.00	0.68				
(9) Slack resource	4.53	0.59	0.07	0.08	0.05	-0.02	-0.04	-0.01	0.06	-0.06	0.71			
(10) Product development alliances	4.64	0.56	0.06	0.00	0.02	0.02	-0.02	-0.13	0.03	0.10	0.03	0.70		
(11) Venture size (log)	5.93	0.59	0.04	0.02	0.13	-0.04	-0.04	-0.01	0.01	0.00	-0.02	0.04	n.a.	
(12) Ownership			0.11	-0.04	-0.12	0.01	0.00	-0.04	0.04	-0.08	0.04	0.05	-0.05	n.a.
Composite reliability (CR)	0.65	0.48	0.78	0.72	0.81	0.71	0.72	0.83	0.82	0.83	0.85	0.85	n.a.	n.a.

Note: The diagonal elements are square roots of the AVE. The lower left triangle elements are correlations analyzed from $N = 227$.

4. Analysis and Results

4.1 *The Links Among Boundary-Spanning Search, Environmental Forces and Breakthrough Innovation*

To test the hypotheses, we employed a path model with the maximum likelihood estimation method. In addition, we included four control variables in this model and linked them directly to boundary-spanning search and breakthrough innovation. Three types of environmental forces—technology turbulence, market turbulence and competitive intensity—have positive effects on supply-side search ($\beta = 0.14$, $p < 0.05$), demand-side search ($\beta = 0.16$, $p < 0.05$) and spatial-side search ($\beta = 0.14$, $p < 0.05$), respectively, in support of H1, H2 and H3. The three types of boundary-spanning search have significant influence on tech-based and market-based innovations. In terms of supply-side search, there was a positive effect on tech-based innovation ($\beta = 0.28$, $p < 0.001$) and a negative effect on market-based innovation ($\beta = -0.14$, $p < 0.05$), in support of H4. Demand-side search has had positive associations with tech-based ($\beta = 0.14$, $p < 0.05$) and market-based ($\beta = 0.24$, $p < 0.001$) innovations, in support of H5a and H5b. Finally, spatial-side search also has had positive impacts on both tech-based ($\beta = 0.22$, $p < 0.001$) and market-based ($\beta = 0.19$, $p < 0.01$) innovations to support H6a and H6b.

4.2 *The Moderating Effects of Boundary-Spanning Search*

We tested the hypotheses with two hierarchical moderated regression analyses for tech-based and market-based innovations. In our study, the relevant variables were mean-centered before creating the interaction terms. Since there were no variance inflation factors in the regression models, above 3 and all were under the threshold of 10; multicollinearity was not a serious issue. Both Models 2 and 5 added the main effects of three environmental situations and three strategic searches and contributed 0.22% ($\Delta F = 9.01$, $p < 0.001$) and 15.7% ($\Delta F = 6.78$, $p < 0.001$) to the variance explained by the control variables. Models 3 and 6 added the interaction terms of three environmental situations with three strategic searches. The increase in these variables explained the variance of 4% ($\Delta F = 3.76$, $p < 0.05$) and 5.6% ($\Delta F = 5.11$, $p < 0.001$) over the explained variance obtained in Models 2 and 5, respectively. In light of H7, it was predicted that the interaction of technology turbulence and supply-side search has had a positive impact on tech-based and market-based innovations. The results showed that the coefficients for interaction term was positive and significant in both of Model 3 ($\beta = 0.15$, $p < 0.01$) and Model 6 ($\beta = 0.20$, $p < 0.01$); hence, H7 was fully supported. Figure 2(a),(b) showed plots of these interactions which provide support for this hypothesis. H8 predicted that the interaction of demand dynamism and demand-side search was positively related to tech-based and market-based innovations in new ventures, respectively. The results demonstrated that the coefficient for interaction term was positive and significant in Model 6 ($\beta = 0.13$, $p < 0.01$) but not in Model 3 ($\beta = -0.04$, $p > 0.10$); thus, H8a was supported. The plot of this interaction shown in Figure 2(c) provides support for this finding. Finally, H9 predicted that the interaction of competitor intensity and spatial-side search has had positive and significant effects on tech-based and market-based innovations. The results revealed that the coefficient for interaction term was positive and significant in Model 3 ($\beta = 0.12$, $p < 0.10$) but not in Model 6

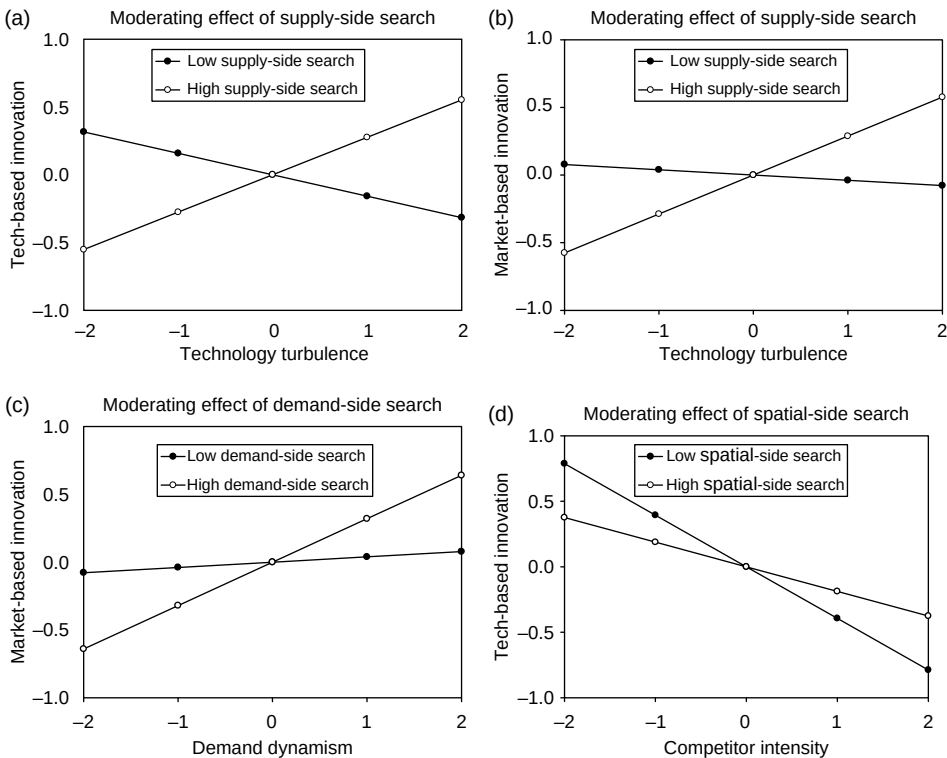


Figure 2. Graphs of the moderating effects.

($\beta = -0.06$, $p > 0.10$); thus, H9a was supported. Figure 2(d) shows the plot of this interaction which provides support for this hypothesis (Table 2).

4.3 Additional Analysis

Our research model may posit that boundary-spanning search mediate the effects of market forces on breakthrough innovations. This study used Baron and Kenny's (1986) tests of mediation to verify this claim. With the entry of three types of boundary-spanning search, the effect of competitive intensity on tech-based innovations ($\beta = 0.24$, $p < 0.001$) is reduced (but remains significant): competitive intensity ($\beta = 0.23$, $p < 0.001$). This suggests partial mediation. Similarly, with the entry of boundary-spanning search, the effect of market turbulence on market-based innovations ($\beta = 0.19$, $p < 0.01$) is reduced but remains significant: market turbulence ($\beta = 0.18$, $p < 0.01$), in support of partial mediation. In addition, our study also examines the effects of different sources of market forces on three types of boundary-spanning search. The results showed that only market turbulence can significantly influence boundary-spanning supply search ($\beta = 0.13$, $t = 2.33$, $p < 0.05$). Table 3 summarizes the hypotheses and empirical conclusions of the study.

Table 2. Results of hierarchical moderated regression analysis.

	Tech-based innovation			Market-based innovation		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Control variables						
Slack resource	0.07 (1.06)	0.06 (1.02)	0.06 (1.14)	0.08 (1.22)	0.09 (1.55)	0.08 (1.31)
Product development alliances	0.05 (0.75)	0.03 (0.51)	0.03 (0.58)	0.00 (0.03)	0.01 (0.20)	−0.00 (−0.01)
Venture size (log)	0.05 (0.75)	0.05 (0.81)	0.05 (0.79)	0.02 (0.31)	0.01 (0.10)	−0.01 (−0.21)
Ownership	0.11 (1.67)	0.11* (2.01)	0.10 (1.78)	−0.04 (−0.55)	0.01 (0.12)	0.01 (0.14)
Independent variables						
Demand dynamism (DD)		−0.04 (−0.64)	−0.04 (−0.77)		0.18* (2.39)	−0.13* (−2.10)
Technology turbulence (TT)		0.04 (0.66)	0.02 (0.33)		0.05 (0.91)	0.22*** (3.71)
Competitor intensity (CI)		−0.23*** (−3.68)	−0.26*** (−4.26)		0.04 (0.65)	0.20*** (3.43)
Demand-side search (DSS)		0.14* (2.20)	0.16* (2.41)		0.24*** (3.86)	0.06 (0.98)
Supply-side search (SSS)		0.28*** (4.71)	0.27*** (4.59)		−0.14* (−2.27)	0.20** (2.75)
Spatial-side search (SPS)		0.22** (3.24)	0.20** (3.09)		0.19** (3.14)	0.03 (0.54)
Moderating effects						
DD × DSS (H7a and H7b)			−0.04 (−0.59)			0.13* (2.09)
TT × SSS (H8a and H8b)			0.15** (3.07)			0.20** (3.30)
CI × SPS (H9a and H9b)			0.12† (1.85)			−0.06 (−1.21)
R^2	0.02	0.22	0.26	0.01	0.16	0.22
Adjusted R^2		0.18	0.21		0.13	0.17
ΔR^2		0.20***	0.04**		0.16***	0.06**
F value	1.20	5.99***	5.65***	0.45	4.28***	4.656***

Notes: This report shows standardized regression coefficients (t value is in parentheses). The t value results from robust standard error. The t tests show one tailed for hypothesized effects and two tailed for controls.

† $p < 0.10$, * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$.

5. Conclusion

This research adopted the environmental strategy coalignment perspective to prove that new ventures in competitive environments may acquire better innovation performance if they align knowledge search strategies with their specific environment (Venkatraman & Prescott, 1990; Lukas *et al.*, 2001). We propose an explanative research framework to

Table 3. Summary of hypotheses and empirical conclusions.

Hypotheses	Expected sign	Empirical conclusions
H1: Technological turbulence is positively related to supply-side boundary-spanning search.	+	Supported
H2: Market turbulence is positively related to demand-side boundary-spanning search.	+	Supported
H3: Competitive intensity is positively related to spatial boundary-spanning search.	+	Supported
H4: Supply-side boundary-spanning search is (a) positively related to tech-based innovations and (b) negatively related to market-based innovations.	+	Supported
H5: Demand-side boundary-spanning search is positively related to (a) tech-based innovations or (b) market-based innovations.	– +	Not supported Supported
H6: Spatial boundary-spanning search is positively related to (a) tech-based innovations and (b) market-based innovations.	+	Supported
H7: The effect of technological turbulence on (a) tech-based innovation is greater when supply-side search is higher than when it is lower and (b) market-based innovation is greater when supply-side search is higher than when it is lower.	+	Supported
H8: The effect of market turbulence on (a) tech-based innovation is greater when supply-side search is higher than when it is lower and (b) market-based innovation is greater when supply-side search is higher than when it is lower.	+	Not supported
H9: The effect of competitive intensity on (a) tech-based innovation is weaker when supply-side search is higher than when it is lower and (b) market-based innovation is greater when supply-side search is higher than when it is lower.	+	Supported
	+	Not supported

probe into the effect of environmental dynamics on boundary-spanning search behavior, the influence of boundary-spanning search behavior on breakthrough innovation and how boundary-spanning search influences the link between market forces and breakthrough innovation. This study expands the literature on knowledge search and innovation search (Rosenkopf & Nerkar, 2001; Katial, 2002; Katila & Chen, 2008) to explore whether the different types of innovation search enable ventures to develop breakthrough innovations, which few scholars have done before as we have done in our work (Katila, 2002; Phene *et al.*, 2006). In addition, we examined the influence among various environmental dimensions, for which no previous theoretical connections have been done until now (Katila & Chen, 2008).

One of our essential arguments is that a venture needs to align its innovation search strategies with the requirements of its environment to achieve breakthrough innovations (Venkatraman & Prescott, 1990; Zajac *et al.*, 2000) as early studies have agreed that a certain market force has both specific constraints and opportunities (Hrebiniak & Joyce, 1985; Sidhu *et al.*, 2007). Accordingly, this study expects that ventures need to engage in appropriate boundary-spanning search strategies in order to deal with different sources of market forces. Our results show that the appropriate search strategies can capture the opportunities and eliminate the threats of different market forces, which in turn can lead ventures to have better innovation performances.

First, when facing comparatively more technological turbulence, like in China, new ventures may need to engage in supply-side search to increase technological reinforcement in product innovation (Katila & Ahuja, 2002; Phene *et al.*, 2006). It implies that adopting supply-side search can accelerate the speed in advancing product innovativeness to rapidly respond to market forces. However, the results show that ventures engaging in supply-side search only enhance their tech-based innovations rather than their market-based innovations. A rational explanation is that supply-side search only emphasizes technology-oriented external knowledge, not the exploration of new market territory and not the reflection of market requirements in a timely manner (Christensen, 1997).

Second, to deal with market turbulence, new ventures should adopt boundary-spanning demand search to capture additional market opportunities and niches (Atuahene-Gima *et al.*, 2005; Sidhu *et al.*, 2007). Our findings show that the demand uncertainty encourages ventures to engage in demand-side search, which facilitates them to develop both tech-based and market-based innovations (Zhou *et al.*, 2005). The implication is that as ventures face market turbulence, adopting demand-side search not only helps them create new segment value but also facilitates to secure the trajectory of future technology (Benner & Tushman, 2003).

Third, under conditions of high competitive intensity, ventures are more likely to adopt boundary-spanning spatial search, which facilitates recombining old external knowledge from outbound industries or international regions to attain competitive advantages. The knowledge acquired from a distant place or from a different kind of industry is increasingly recognized to be novel and useful for new ventures to develop breakthrough innovations (Katila, 2002; Phene *et al.*, 2006). Our findings show that adopting spatial search facilitates increasing consumer knowledge and relatedness outside of the market territory to develop market-based innovations. The results also reveal that this external search enables ventures to improve production technology through the good use of the best module of prior

practice experiences and then turning this improved production technology into valuable innovations (Kogut & Zander, 1992; Henderson & Cockburn, 1994).

Finally, the results highlight that boundary-spanning search may moderate the effects of market forces on breakthrough innovations. This contribution is important because the moderating effects refine the essence of the environmental strategy coalignment perspective. For example, new ventures located in an environment of high-technological turbulence, like in China, can have better tech-based innovation performance if they learn to engage in more supply-side search. Moreover, when new ventures face high market turbulence, they need to engage in more demand-side search because the search enables them to achieve tech-based and market-based innovations simultaneously. Furthermore, ventures should be involved in more spatial-side search under conditions of high-competitive intensity if they expect to secure higher breakthrough innovations. Consequently, for survival in competitive environments, new ventures should pay more attention to coalign market forces and innovation search strategies to attain breakthrough innovations.

5.1 Limitations and Future Research Directions

There are some limitations to our research. First, although this paper is a causal study to investigate the relationships among market forces, boundary-spanning searches and breakthrough innovations, it is fundamentally a cross-sectional study using the survey method so that the cause and effect of our finding is not directly measured. Future research can employ a longitudinal study to more exactly investigate the relationships established in this research. Second, our sample focuses on new ventures in high-technological industries. In order to generalize our research results, it is recommended to examine our findings for each of the different industries and cultures. Third, our research hypothesizes that new ventures could use specific types of boundary-spanning search in accordance with specific environments. Therefore, it is suggested that future research can be more detailed in the investigation of the relationships between market forces and boundary-spanning searches. Finally, this paper adopts two breakthrough innovations as our research outcomes, such that the notion of these innovations includes space, target customers and the speed of technology. However, our research investigates only the general concepts of these two breakthrough innovations. Future research should more deeply re-examine innovations in accordance with these three dimensions.

5.2 Suggestions

Based on the logic of the environmental strategy coalignment perspective (Venkatraman & Prescott, 1990), we have found that different environmental forces encourage venture managers to adopt different external knowledge search strategies (boundary-spanning search) for developing better innovation performance. Specifically, our results remind the reader that boundary-spanning search strategies can strengthen new venture's response to the effect of market forces on developing breakthrough innovations. In other words, new ventures in competitive environments are capable of managing the influence of market forces on innovative outcomes when they can better engage in more boundary-spanning search strategies (Clark *et al.*, 1994; Li, 2001).

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Appendix: Measures and Confirmatory Factor Analysis Results

Technology turbulence^a (Jaworski & Kohli, 1993)

With respect to your industry's environment:

- 1 The technology in the market environment was changing rapidly.
- 2 Technological changes provide big opportunities in the industry.
- 3 A large number of new product ideas have been made possible through technological breakthroughs in our industry.
- 4 There are major technological developments in the industry.

Market turbulence^a (Jaworski & Kohli, 1993)

With respect to your industry's environment:

- 1 Customers in our industry tend to look for new products all the time.
- 2 The product-market competitive conditions are highly unpredictable.
- 3 Customers' preferences change quite rapidly.
- 4 Customers' needs in our industry are changing quite rapidly.

Competitive intensity^a (Yang & Li, 2011)

With respect to your industry's environment:

- 1 Competition in our market is intense.
- 2 Our organization has relatively strong competitors.
- 3 Competition in our market is extremely high.
- 4 Price competition is a hallmark of our market.

Tech-based innovations^a (Zhou *et al.*, 2005)

- 1 Our product is highly innovative, replacing an inferior alternative.
- 2 Our product incorporates a radically new technological knowledge.
- 3 Overall, our product is similar to our main competitors' products. (R)^d
- 4 The application of our product is totally different from that of our main competitors' products.

Market-based innovations^a (Zhou *et al.*, 2005)

- 1 Our product concept is difficult for mainstream customers to evaluate or understand.
- 2 Our product involves high switching costs for mainstream customers.
- 3 The use of our product requires a major learning effort by mainstream customers.
- 4 It takes a long time for mainstream customers to understand our product's full benefits.

Supply-side search^b (Sidhu *et al.*, 2007)

Over the last 3 years,

- 1 We are well aware of technological and technical developments within our industry.
- 2 Our information-gathering efforts cover all industries that employ that sort of technology that we use.
- 3 A careful watch is kept on industries that are technologically related to ours (e.g., telecom and computer industries are technologically related).
- 4 We acquire little information on opportunities to employ our existing production facilities in new product domains. (R)
- 5 We closely monitor companies not active in our product area, but that have skills and know-how comparable to ours.
- 6 In our company, there is close surveillance of advancements in product and process technologies in supplier industries.

Demand-side search^b (Sidhu *et al.*, 2007)

Over the last 3 years,

- 1 Marketing strategies of companies targeting our customers are closely followed by us.
- 2 We have a finger on the pulse as far as changes in the product preferences of our customers are concerned.
- 3 Little information is gathered on product preferences of customer groups that we do not currently serve. (R)

4 Developments in industries that fulfill the same customer need as we do, albeit with a completely different product, are well known to us (e.g., air and train transport both fulfill customer need for mobility).

5 We keep close track of activities of companies that offer complementary products (e.g., cameras and film rolls are complementary products because they are used together by customers).

6 We know the product and process innovation efforts of our customers well.

Spatial search^b (Sidhu *et al.*, 2007)

Over the last 3 years,

1 We are knowledgeable about all important opportunities in the geographic regions in which we operate.

2 We hardly acquire any intelligence about potential opportunities in new geographic markets. (R)

3 We are well informed about the price and quality aspects of products in neighboring geographic regions.

4 We closely follow the activities of companies in our industrial sector but operating outside our geographic area.

Slack resources^a (De Luca & Atuahene-Gima, 2007)

1 We have a large amount of resources available in the short run to fund our initiatives.

2 We have uncommitted resources that can be used to fund strategic initiatives at short notice.

3 We have a lot of resources at the discretion of management to fund new strategic initiatives.

4 We are able to obtain resources at short notice to support new strategic initiatives.

Product development alliances^c (Li & Atuahene-Gima, 2002)

Over the last 3 years?

1 Established cooperative R&D agreements with other firms.

2 Jointly introduced new products to market with other firms.

3 Jointly designed and manufactured new products with other firms.

4 Marketed complementary new products with other firms.

5 Jointly distributed and provided new product support services with other firms.

6 Promoted new product lines jointly with other firms.

^a The items were measured on a seven-point scale, anchored by 1 = strongly disagree and 7 = strongly agree.

^b The items were measure on a seven-point scale, anchored by 1 = no extent and 7 = to a great extent.

^c The items were measure on a seven-point scale, anchored by 1 = worse than major competitor and 7 = far better than major competitor.

^d (R) = a reverse-worded item.